ORIGINAL ARTICLE

Kinematic and Clinical Analyses of Upper-Extremity Movements After Constraint-Induced Movement Therapy in Patients With Stroke: A Randomized Controlled Trial

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ABSTRACT. Wu CY, Chen CL, Tang SF, Lin KC, Huang YY. Kinematic and clinical analyses of upper-extremity movements after constraint-induced movement therapy in patients with stroke: a randomized controlled trial. Arch Phys Med Rehabil 2007;88:964-70.

Objective: To study the effects of constraint-induced movement therapy (CIMT) relative to traditional intervention on motor-control strategies for upper-arm reaching and motor performance at the impairment and functional levels in stroke patients.

Design: Two-group randomized controlled trial (RCT); pretreatment and posttreatment measures.

Setting: Rehabilitation clinics.

Participants: Forty-seven stroke patients (mean age, 55y) 3 weeks to 37 months postonset of a first-ever cerebrovascular accident.

Interventions: Forty-seven patients received either CIMT (restraint of the less affected hand combined with intensive training of the more affected upper extremity) or traditional intervention (control treatment) during the study. The treatment intensity was matched between the 2 groups (2h/d, 5d/wk for 3wk).

Main Outcome Measures: Outcomes were evaluated using (1) kinematic variables of reaching movement used to describe the control strategies for reaching, (2) the Fugl-Meyer Assessment (FMA) of motor-impairment severity, and (3) the Motor Activity Log (MAL) evaluating the functional ability of the upper extremity.

Results: After treatment, the CIMT group showed better strategies of reaching control than the control group ($P<.03$). The CIMT group also showed less motor impairment on the FMA ($P=.019$) and higher functional ability on the MAL ($P<.001$).

Conclusions: This study is the first RCT to show differences in motor-control strategies as measured by kinematic variables after CIMT versus traditional intervention. In addition to improving motor performance at the impairment and functional levels, CIMT conferred therapeutic benefits on control strategies determined by kinematic analysis.

Key Words: Controlled clinical trials; Kinematics; Occupational therapy; Rehabilitation; Stroke.

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WITH THE PROLIFERATION OF successful controlled clinical trials of constraint-induced movement therapy (CIMT) in the literature, the therapy has attracted considerable attention as a means to treat the more affected upper extremity (UE) and overcome the learned nonuse phenomenon (habitually relying on the less affected UE to accomplish functional tasks) among patients with stroke.[1-5] CIMT involves restraint of the less affected UE over an extended period (6–18h/d for 2–3wk), in combination with intensive task-specific training of the more affected limb (eg, 6h/d on 10–15 consecutive weekdays).[4] In contrast to clinical efficacy of CIMT, the acceptance of CIMT among therapists and patients remains poor. One possible reason is that intense and massed practice during CIMT may be less safe and more tiring for stroke patients.[6] To address the problems, a variety of derivatives of CIMT were devised by using less intensive training and shorter restraint time. These derivatives of CIMT may involve training of the affected UE for 0.5 to 3 hours a day, 3 to 5 days a week for 2 to 10 weeks, together with restraint of the less affected UE for up to 6 hours a day.[7-14] Substantial evidence shows that CIMT can improve motor impairment and functional use of the more affected UE in chronic (>1y poststroke),[8,9] subacute (>3mo <1y poststroke),[7,10,11] and acute[10,12-14] stroke patients with mild motor impairments. However, the motor-control mechanisms responsible for the improved motor performance are poorly understood. A possible reason for this lack of understanding relates to the methods used to assess UE function.[15]

Most studies of CIMT have relied on clinical evaluations such as observer-initiated measure of motor impairment (eg, the Fugl-Meyer Assessment [FMA]) or self-report measure of functional ability (eg, Motor Activity Log [MAL]).[7-9,11-14] These tests provide clinical information regarding the level of motor impairment and the level of functional ability or patients' perceptions of how well they can functionally use their UEs. However, these measures may have particular aspects that are left up to the subjective perspective of the rater.[12,15] Technologic developments in motion analysis enabled biomechanics studies of actions (eg, upper-arm reaching). Biomechanic analysis of movements provides more objective and quantitative measures of control strategies necessary for promoting understanding of the mechanisms underlying improved motor performance of

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stroke patients after CIMT.[16] Recent research has recommended the use of biomechanic analysis for outcome evaluations of CIMT.[12]

Alberts et al[15] used kinetic analysis to objectively examine how CIMT affects control of force in precision grip. In addition to kinetic analysis, kinematic analysis during functional tasks has been suggested as a valid means to directly and objectively measure the spatiotemporal control of movement.[17] Kinematic measures of reaching have been shown to discriminate changes to hand-path quality after brain injury[18] and are associated with measures of arm functions (eg, the Action Research Arm Test).[19] Such measures can be used not only to assess performance but also to identify movement quality or to elucidate the motor strategies during a reaching task.[17] The information may afford insights into how motor-control rehabilitation alters movement organization in patients with stroke.

A normal-reaching movement is controlled by both preprogramming and online error correction.[20-22] A more preprogrammed control strategy for reaching movement represents a better learned or more skillful movement. When the movement depends more on motor preprogramming, it will be more rapidly initiated, more efficient and direct, and smoother. Preprogrammed movement requires rapid task completion, such that the performer may need greater force or impulse at movement initiation to quickly bring the hand to the target. In contrast, if the control strategy of the movement depends more on online correction, it will require more planning and therefore be initiated more slowly. This type of movement requires adjustment of direction based on sensory feedback during performance, resulting in reduced movement efficiency, straightness, and smoothness.

A recent case report[23] has used kinematic measures to evaluate the changes in motor control after CIMT and presented preliminary evidence that more preprogrammed control strategy of reaching movement is achieved after than before CIMT. Further research performed by using a randomized controlled trial (RCT) is needed to afford insights into mechanisms mediating change in motor performance after CIMT. The present study used kinematic analysis to investigate the impact of CIMT versus traditional intervention on control strategies for reaching movement and on motor impairment and functional ability of the UE after stroke. Combining kinematic analysis and clinical evaluation might enable comprehensive assessment of the change in control strategies and motor performance at the impairment and functional levels after CIMT. Reaching was selected as the study task because of its status as a fundamental motor skill necessary for many daily activities. We hypothesized that patients receiving 3 weeks of CIMT would exhibit better strategies of reaching control, reduced motor impairment, and increased functional ability involving the more affected UE than patients receiving the traditional intervention. Better control strategies would be evidenced by more efficient, straighter, and smoother movements. Reduced motor impairment and increased functional ability would be represented by higher scores of FMA and MAL.

METHODS

Participants

The institutional review boards for human studies at 2 medical centers approved this protocol, and all subjects gave informed consent. Subjects were recruited from 2 stroke rehabilitation units; 47 stroke patients (32 men, 15 women; mean age, 55y; range, 40–80y) participated. According to self-report, subjects were right-hand dominant before stroke. At the beginning of the intervention, they were 3 weeks to 37 months (mean, 12.25mo) postonset of a first-time cerebrovascular accident of ischemic or hemorrhagic type. Inclusion criteria were as follows: (1) able to reach Brunnstrom stage[24] III or above for the proximal part of the UE, (2) considerable nonuse of the more affected UE (MAL amount of use [AOU][25] score <2.5), (3) no serious cognitive deficits (modified Mini-Mental State Examination[26] score ≥ 70), (4) no balance problems sufficient to compromise safety when wearing the experimental constraint device, and (5) lack of participation in any experimental rehabilitation or drug studies and absence of use of antispasticity drugs for UE musculature (eg, botulinum toxin type A) within the past 3 months. All patients received independent examinations by a physiatrist and an occupational therapist to determine their eligibility for inclusion. Figure 1 details subject recruitment and assignment.

Instruments

Movement data were collected using kinematic analysis and clinical evaluation as described later.

Kinematic analysis. A desk bell (diameter, 9.7cm [3.80in]; height, 4.8cm [1.87in]) was used as the target for reaching to press it. During reaching, a 6-camera motion analysis system[a] was used in conjunction with a personal computer to capture the movement of a marker attached to the styloid process of the ulna; 2 channels of analog signals were collected simultaneously. The analog signals were connected to a pressure-sensitive hand switch and the desk bell. Movement recording began when the hand moved off the hand switch, and movement termination was recorded when the subject pressed the desk bell. Movements were recorded at 60Hz and digitally low-pass filtered at 5Hz using a second-order Butterworth filter with a forward and backward pass.

Clinical evaluation. The clinical measures used in this study included the FMA and the MAL. The FMA was used to assess several dimensions of motor impairment. The MAL is a

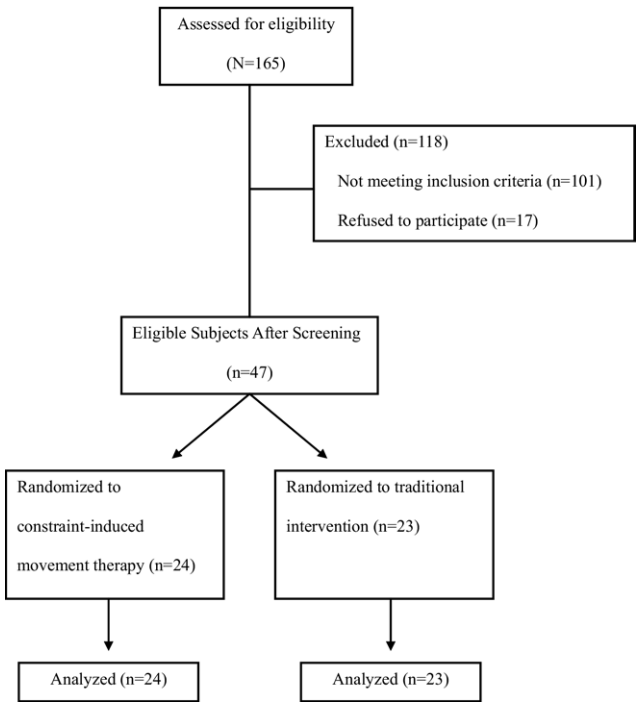


Fig 1. Flowchart of subject recruitment and group assignment.

functional measure and used to evaluate the ability of performing essential tasks and functional activities through self-report.

Scoring on the arm and hand section of the FMA (maximum score, 66) was based on a 3-point ordinal scale (0, cannot perform; 1, can perform partially; 2, can perform fully).[27] A higher FMA score indicates less motor impairment. Test-retest reliability, interrater reliability, and construct validity of the test are well established.[28,29]

The MAL is a semistructured interview that obtained information about how patients use their more affected limbs during 30 important daily activities. Patients were instructed to use a 6-point AOU scale (score range, 0–5) to rate how much the arm is being used and a 6-point quality of movement (QOM) scale (score range, 0–5) to rate how well they are using their more affected UEs during the past week.[25]

Intervention

Treatment regimens were designed to ensure that patients received equal treatment intensity (2h/d, 5d/wk for 3 consecutive weeks) directly supervised by the occupational therapists. The intervention was provided at 2 centers under the supervision of 2 separate occupational therapists. These 2 therapists were trained in the administration of the CIMT protocol by the investigators and completed a written competency test before subject treatment. During the treatment period, structured daily treatment notes were made and reviewed by the investigators to ensure the standardization of treatment. All trainings were provided on an individual basis. Subjects were blind to the study hypotheses. All subjects received routine interdisciplinary stroke rehabilitation separate from the study treatment that occurred during the regularly scheduled occupational therapy sessions. The interdisciplinary stroke rehabilitation was delivered by a variety of treatment disciplines including physical therapists and psychologists. The intensity of the interdisciplinary rehabilitation was the same for all participants (1.5h/d for 5d/wk).

In the CIMT group, typical training activities involved the use of the more affected UE and were similar to those performed daily (eg, reaching forward to move a jar from 1 place to another, picking up a cup and drinking from it, picking up a hairbrush and combing hair, cleaning the window). The less affected hand was placed in a mitt for 6 hours a day throughout the study period. In the traditional intervention group, the treatment involved neurodevelopmental therapy emphasizing functional task practice when possible, stretching and weight bearing with the more affected arm, and fine-motor dexterity training.

Procedures and Testing

Subjects were randomly assigned to the CIMT or traditional intervention group by using a random numbers table. Before and after the 3-week intervention period, the laboratory test (ie, kinematic analysis) and the clinical evaluation (FMA, MAL) were administered in random order by a blinded rater. The order of the laboratory test and the clinical evaluation was randomized to wash out carryover effects. Before administration of the outcome measures, a blinded rater was trained to administer the FMA and MAL following the guidelines described by Fugl-Meyer[27] and Taub[30] and colleagues, respectively. Rater competence was assessed by the primary investigators who had 5 years of experience in using these measures. The rater was trained to conduct the kinematic analysis in accord with standardized procedures described as follows.

During the laboratory test for reaching kinematics, each subject sat on a height-adjustable chair with seat-height set to 100% of the lower leg length, measured from the lateral knee joint to the floor with the subject standing. The trunk was secured to the chair back with a harness to prevent lateral and forward flexion and rotation but still allow for scapular motion. The subject rested the more affected hand on the hand switch placed on the edge of the table in line with the subject's midsagittal plane. Table height was adjusted to 5.1cm (2in) below the elbow. The desk bell was located along the subject's midsagittal plane, and the reaching distance to the bell was standardized to the subject's functional arm length. Functional arm length was defined as the distance from the medial border of the axilla to the distal wrist crease when the subject raised his/her arm as close to 90° elevation as possible and reached forward (without trunk movement) as far as possible. The functional arm length represents the farthest distance the subject can reach forward without using trunk movement. Subjects were instructed to reach and press the desk bell using the hand at a comfortable speed; 5 trials were performed after a practice trial.

Data Reduction and Data Analysis

An analysis program coded by LabView[b] language was used to process the kinematic data. Values of reaction time, movement time, total displacement, peak velocity, and movement units were obtained. Because the task distance varied across subjects, movement time, total displacement, and movement units were normalized to correct for variations in reaching distance.

Reaction time refers to the time to initiate the movement. Movement time means the time for execution of the reaching movement, representing movement efficiency. Total displacement refers to the path of the hand in 3-dimensional space, indicating the directness of movement. One movement unit is comprised of 1 acceleration and 1 deceleration phase and can be used to characterize movement smoothness and evaluate the extent of error correction during movement performance. Fewer movement units indicate smoother movement.[31,32] Peak velocity indicates force or impulse at movement initiation. Greater force or impulse is reflected in a higher-amplitude peak velocity.

Analysis of covariance (ANCOVA),[33] controlling for pretreatment differences, was used to test whether the CIMT group performed significantly better than the traditional intervention group on the posttest for each variable. For each analysis, pretest performance (kinematic data, FMA and MAL scores) and time postonset of stroke were used as the covariates, group was the independent variable, and posttest performance was the dependent variable. The effect size r was calculated for each outcome variable to index the magnitude of the performance difference between groups.[34] According to Cohen,[35] a large effect is represented by an r of at least .50, a moderate effect by .30, and a small effect by .10.

RESULTS

There were no significant differences between the 2 groups with respect to the measured characteristics except for the pretreatment kinematic performance on normalized total displacement (table 1). We addressed the pretest variability between groups by treating the pretreatment performance as a covariate in the comparison of posttest performance by the 2 groups.

The ANCOVA results showed significant and moderate effects on most kinematic measures at posttest (table 2). Patients treated with CIMT were able to initiate movement more quickly than patients treated with traditional intervention, as

Table 1: Subject Clinical and Demographic Characteristics

Characteristics	CIMT (n=24)	TI (n=23)	P*
Sex (male/female)	16/8	17/7	.83
Mean age ± SD (y)	53.93±11.20	56.77±12.90	.42
Side of lesion (right/left)	11/13	11/12	.89
Mean months after stroke ± SD	12.51±9.64	11.98±11.72	.87
Between 0.6 and 6m (n)	9	10	
Between 6 and 12m (n)	5	4	
Between 12 and 24m (n)	6	6	
Between 24 and 37m (n)	4	3	
Brunnstrom stage (proximal part of UE) (median)	4.5	4.5	.39
Mean modified MMSE score ± SD	84.42±10.35	84.43±11.62	.99
Mean pretreatment performance on kinematic variables ± SD			
Reaction time (s)	0.72±0.46	0.60±0.28	.28
Normalized movement time	0.07±0.05	0.09±0.06	.95
Normalized total displacement	1.70±0.51	1.42±0.36	.036
Peak velocity (cm/s)	66.32±22.88	55.35±15.45	.062
Normalized movement units	7.06±5.26	5.69±6.25	.42
Mean pretreatment scores on FMA ± SD			
UE	39.50±13.45	41.74±13.47	.64
Mean pretreatment scores on MAL ± SD			
AOU	0.64±0.86	0.60±0.92	.87
QOM	0.72±1.01	0.69±1.17	.94

Abbreviations: MMSE, Mini-Mental State Examination; SD, standard deviation; TI, traditional intervention.
*P associated with the chi-square test for categorical variables, the independent *t* test for continuous variables, and the Mann-Whitney *U* test for the ordinal variable.

evidenced by a shorter reaction time ($P=.005$). The CIMT group performed the task more efficiently (shorter normalized movement time, $P=.021$) and with straighter (smaller normalized total displacement, $P=.021$) and smoother (fewer normalized movement units, $P=.010$) reaching trajectories of the more affected arm. Thus, the CIMT group showed more preprogrammed movement after treatment than did the traditional intervention group. A nonsignificant and small effect was found for the kinematic variable peak velocity.

The results showed significant and moderate to large effects in favor of the CIMT group on the FMA and MAL. There was less motor impairment for the CIMT group versus the traditional intervention group (FMA, $P=.019$). Subjects in the CIMT group reported better performance in AOU ($P<.001$) and QOM ($P<.001$) of their more affected UE during daily activities. The CIMT group reported using the more affected UE for an average of 14 activities and the traditional intervention group for an average of 15 activities before treatment and both groups for 24 after treatment.

DISCUSSION

This study is, to our knowledge, the first RCT to use kinematic analysis to investigate differences in motor-control strategies after CIMT versus traditional intervention. We observed better performance in reaching kinematics of the UE after CIMT as compared with the traditional intervention. The CIMT group also showed less motor impairment assessed by the FMA and greater functional ability evaluated by the MAL. The beneficial effects of CIMT are consistent with previous findings,[7-9,11-13] but the findings further suggest that CIMT may improve motor-control strategies evidenced by kinematic data.

The kinematic results of this study extend those of the case report by Hakim et al[23] to a larger sample. The enhanced

Table 2: Descriptive and Inferential Statistics for Analysis of Reaching Kinematics and Clinical Assessment

	Posttreatment		ANCOVA		
Assessment	CIMT (n=24)	TI (n=23)	$F_{1,44}$	*P*	Effect Size *r*
Kinematic variables					
Reaction time (s)	0.48±0.17	0.63±0.32	8.86	.005	.41
Normalized movement time	0.04±0.03	0.05±0.03	5.72	.021	.34
Normalized total displacement	1.32±0.22	1.42±0.39	5.75	.021	.34
Peak velocity (cm/s)	76.42±16.17	65.72±18.79	1.10	.30	.16
Normalized movement units	0.13±0.11	0.19±0.16	7.34	.010	.38
FMA					
UE	46.75±11.58	44.78±13.08	5.97	.019	.35
MAL					
AOU	1.85±1.24	0.81±1.13	32.76	<.001	.66
QOM	1.85±1.14	0.84±1.08	31.59	<.001	.65

performance in reaching kinematics (ie, movement quality) of the CIMT group may reflect better control strategies (ie, a more preprogrammed strategy) as a result of forced use of the more affected UE for intensive practice on functional tasks. The tasks offered motor problems to be solved and allowed subjects to experiment on the solutions to accomplish the task goal. Moreover, the decreased standard deviations for kinematic variables imply a more stable movement pattern in the CIMT group. Such better performance indicates that subjects who had received CIMT were able to perform more preprogrammed movement than those who had received the traditional intervention. In other words, subjects who had received CIMT used their more affected UEs more spontaneously and could modify ongoing movement with less attentional effort. Additional evidence for better movement control after CIMT was evidenced by faster movement planning or preparation (less reaction time), increased movement efficiency (less normalized movement time), and a straighter and smoother movement trajectory (less normalized total displacement and fewer movement units). This enhanced movement control may have related to short-term learning changes at central and spinal levels.[36] Intensive training of the more affected UE may, at the central level, enhance motor planning and at the spinal level decrease the latency between activation of agonist and antagonist muscles, leading to a shorter reaction time.[37] Extensive practice with a variety of functional tasks may have provided opportunity for the patients to experiment with efficient arm use for reaching movements and regain motor skills and thus perform the reaching task faster and more efficiently. Straighter and smoother movements might be caused, at the central level, by improved motor planning involving interjoint coordination.[38] Improved coordination of movements might also, at the spinal level, be caused by an increase in the intensity of activation of spinal motoneuron pools, leading to increased efficiency and coordination of muscular contraction.[36]

The difference in peak velocity posttreatment was not significant between CIMT and traditional intervention. One possibility pertains to the large variability in peak velocity within both groups. Another possibility is that peak velocity is an integral factor involving both spatial and temporal properties of movements and is associated with force or impulse at movement initiation.[32] Because neither treatment program in our study emphasized muscle strengthening or force control, force-related performance during reaching movements may be considered a less sensitive variable for detecting differences in therapeutic effect. When improvement of movement velocity is an endpoint, treatment should incorporate tasks demanding force control (eg, turning a key to open a door). Future research might also evaluate force-related performance during speed-emphasized reaching tasks.

The enhanced performance in movement kinematics associated with CIMT suggests that the therapeutic gains may reflect reduced deficits in motor-control strategies. Movement information on control strategies derived from kinematic analyses may complement outcomes of impairment and functional performance derived from clinical measures and may allow a better understanding of motor-control mechanisms underlying CIMT. This knowledge about how patients with stroke perform reaching movements provides unique perspectives on characteristics of control strategies and may contribute to outcome evaluations in stroke rehabilitation. It is also important for future research to use kinematic analysis to examine motor-control strategies in a variety of functional UE tasks to investigate the generalizability of the current findings to different types of daily activities. Several neuroimaging studies[39-43] have shown that CIMT leads to cortical reorganization with extension, shift, and recruitment of ipsilesional, perilesional, and contralesional cortical areas of the sensorimotor network. It remains unclear whether the therapeutic benefit is attributable to neuroplastic changes in brain areas subserving activity-dependent cortical reorganization. A possibility for future research is to use functional magnetic resonance imaging analysis pre- and post-CIMT to investigate the possible relation between kinematic improvement of reaching movements and altered brain activation.

The finding of better performance represented by FMA and MAL scores after CIMT versus traditional intervention may reflect less motor impairment and increased use of the more affected UE after CIMT. The finding on the FMA is consistent with most previous studies[7,8,12,14] but not that of van der Lee et al.[3] This discrepancy might be primarily caused by differences in the treatment program for the comparison group. The present study, along with those of Page,[7,8,12] and Boake[14] and colleagues, used the traditional intervention for the comparison group, whereas the study by van der Lee[3] used bimanual task training for the comparison group. Bimanual task training involved mass practice of both hands, resulting in only slightly greater reduction of motor impairment in the CIMT versus comparison group. The findings on the MAL are consistent with previous findings[7-9,12,13] and suggest that the learned nonuse phenomenon observed in the patients with stroke can be overcome through CIMT.

This study recruited patients with various levels of arm-movement impairment ranging from hemiplegia with persistent synergy patterns but preservation of some isolated arm movements to hemiparetic weakness with isolated finger movements. The application of CIMT might not be limited to the high-functioning patients having finger movements and might be extended to the patients having only some isolated arm movements. The stage of recovery from stroke in subjects of this study spanned from acute, to subacute, and to chronic. The issue that the natural recovery of stroke might confound the observed effects of CIMT was taken into account in this study. There were almost equal numbers of subjects at each recovery stage between the 2 groups. The variable of time poststroke was further addressed in the statistical analysis by treating it as a covariate. The confounding effect of natural recovery is thus not a plausible explanation for the beneficial effects of CIMT.

Study Limitations

There is a noteworthy limitation of our study. The instruments used in the study assessed control strategies inside the laboratory. Future research might determine if the therapeutic effect of CIMT transfers to the nonlaboratory environment. The absence of electromyographic data in the current study is another limitation; evaluation of muscle-activation patterns together with evaluation of spatiotemporal-movement control might provide a more complete picture of the specific movement parameters that may be affected by CIMT. A clearer understanding of which movement parameters are responsive to therapeutic change and the nature of such change can provide insights into the mechanisms responsible for the therapeutic effects on control strategies. As a further limitation, the long-term effects of CIMT on kinematic performance are unclear and await longitudinal study with long-term follow-up.[44] Future research may also study whether improvements in control strategies after CIMT predict better motor relearning abilities, more recovery in daily functions, and increased potential for return to work.

CONCLUSIONS

This RCT used kinematic and clinical analyses to study postintervention differences between CIMT and traditional intervention in stroke patients varying in the level of arm motor impairment. The study showed CIMT improved both kinematic performance (except for peak velocity) at the level of control strategy and motor performance at the impairment and functional levels. Further research may use brain-imaging techniques to elucidate whether improved control strategies after CIMT are associated with treatment-induced cortical reorganization.

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Suppliers

a. Model 370; Vicon, 14 Minns Business Pk, West Way, Oxford OX2 0JB, UK.
b. National Instruments Corp, 11500 N Mopac Expwy, Austin, TX 78759-3504.